

Decision Tree → classification & Regression

class	Gender	Stay in hostel
9	M	Yes
10	f	No
8	f	Yes
8	f	No
9	M	Yes
10	M	No
11	f	Yes
11	M	Yes
8	f	Yes
9	M	No
11	M	No
11	M	Yes
10	f	No
10	M	Yes

$$Gini = 1 - \sum_{i=1}^c p_i^2$$

$$Gini(class) = 0.404$$

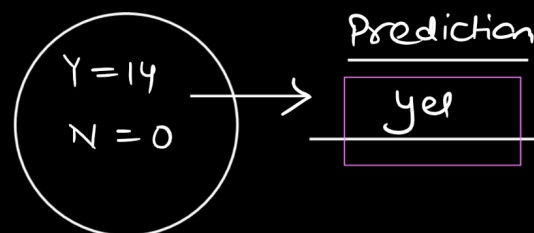
$$Gini(gender) = 0.482$$

↳ 0 to 0.5

↳ Range of Gini

Pure Decision mode

Leaf Node



$$Gini\ index \rightarrow 1 - \sum_{i=1}^c p_i^2$$

$$P(Y) = 14/14 = 1$$

$$P(N) = 0/14 = 0$$

highly pure

$$1 - (1^2 + 0^2) \Rightarrow \underline{\underline{0}} \text{ (Min)}$$

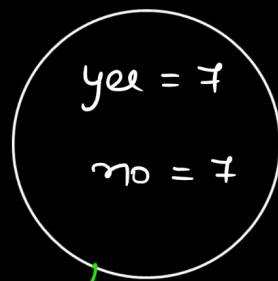
$$\Rightarrow 1 - \left(\frac{1}{2}^2 + \frac{1}{2}^2 \right)$$

$$\Rightarrow 1 - \left(\frac{1}{4} + \frac{1}{4} \right)$$

$$\Rightarrow \underline{0.5}$$

Highly

impure



Prediction

highly Confused
(yes/no)

$$\begin{cases} P(\text{yes}) = \frac{7}{14} = \frac{1}{2} \\ P(\text{no}) = \frac{7}{14} = \frac{1}{2} \end{cases}$$

By default, DT model uses Gini as the

impurity measure

↳ computationally
easier task as

comparable to

compute Entropy

Entropy

$$-\sum_{i=1}^c p_i \log p_i$$

Randomness (chemistry)

(Measure of impurity)

class	Gender	stay in hostel
9	M	Yes
10	F	No
8	F	Yes
8	F	No
9	M	Yes
10	M	No
11	F	Yes
11	M	Yes
8	F	Yes
9	M	No
11	M	No
11	M	Yes
10	F	No
10	M	Yes

class	Stay in hostel	P(Y)	P(N)
8	$y=2 \quad N=1$	$\frac{2}{3}$	$\frac{1}{3}$
9	$y=2 \quad N=1$	$\frac{2}{3}$	$\frac{1}{3}$
10	$y=1 \quad N=3$	$\frac{1}{4}$	$\frac{3}{4}$
11	$y=3 \quad N=1$	$\frac{3}{4}$	$\frac{1}{4}$

$$\text{Entropy (class=8)} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$\Rightarrow \underline{0.918}$$

$$\text{Entropy (class=9)} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$\Rightarrow \underline{0.918}$$

$$\text{Entropy (class=10)} = -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4}$$

$$\Rightarrow \underline{0.811}$$

$$\text{Entropy (class=11)} = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$\Rightarrow \underline{0.811}$$

Entropy (class)

$$\text{Weighted Entropy} \Rightarrow \frac{3}{14} * 0.918 + \frac{3}{14} * 0.918 + \frac{4}{14} * 0.811 +$$

$$\frac{4}{14} \times 0.811$$



$$\underline{\underline{0.857}}$$

Gender	stay in hostel		$P(Y)$	$P(N)$	Entropy
m	y=5	N=3	$5/8$	$3/8$	$-5/8 \log_2 5/8 - 3/8 \log_2 3/8$
f	y=3	N=3	$3/6$	$3/6$	$-1/2 \log_2 1/2 - 1/2 \log_2 1/2$

$\Rightarrow 1$

$$\underline{\text{Entropy(Gender)}} \Rightarrow \frac{8}{14} \times 0.954 + \frac{6}{14} \times 1$$

$$\Rightarrow \underline{\underline{0.975}}$$

Highly

pure

0 to 0.5

highly impure

Gini computation

Entropy computation

→ same results

↳ class

↳ Range of Entropy

$$Y = 14$$

$$N = 0$$

Pure

decision

mode

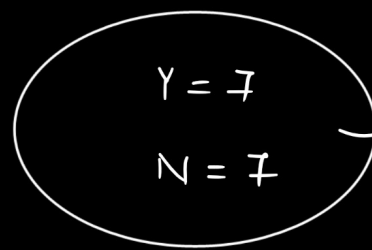
$$P(Y) = 1$$

$$P(N) = 0$$

$$\rightarrow -1 \log_2 1 -$$

$$0 \log_2 0$$

$$\Rightarrow \underline{\underline{0}}$$



Impure

Decision

node

$$P(Y) = \frac{1}{2}, P(N) = \frac{1}{2}$$

$$-\frac{1}{2} \log \frac{1}{2}$$

$$-\frac{1}{2} \log \frac{1}{2}$$

$\Rightarrow$ 1

Major differences b/w Gini vs Entropy?

① Way of computation $\rightarrow$ Gini is faster

or comparable to

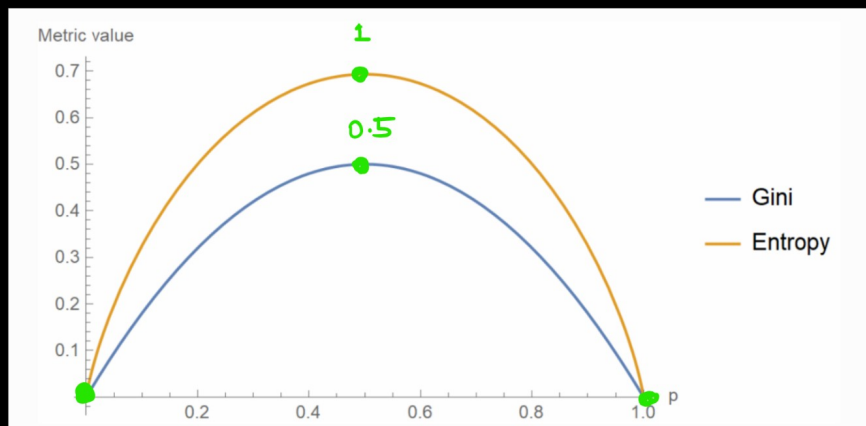
Entropy

②

Range

$\rightarrow$ Gini $\rightarrow$ 0 to 0.5

$\rightarrow$ Entropy $\rightarrow$ 0 to 1



③

by default, DTree documentation
the criterion is Gini

for example $\rightarrow$ Healthcare problem



Entropy Diabetic $\rightarrow 1$ & 40%
can be more non-diabetic $\rightarrow 0$ & 60%
sensitive towards class distribution

- faster computation

(for a very $\Rightarrow$ Gini
huge amount
of data)

Hyperparameter Tuning $\rightarrow$ GridSearchCV
 $\rightarrow$ RandomizedSearchCV

use Gini When

95%

- ↳ faster training
- ↳ very huge amount of data
- ↳ when you need efficient computation
- ↳ good option without doing any
hyperparameter tuning

use entropy when

↳ deeper splits

↳ class distributions are very important
(sensitive problems e.g. healthcare problem)

↳ you want more informative splits

target = stay in hostel

Representation → talk

Loss/error → MSE

$$P(y) \rightarrow -\frac{8}{14} \log_2 \frac{8}{14} - \frac{6}{14} \log_2 \frac{6}{14} \Rightarrow 0.985$$

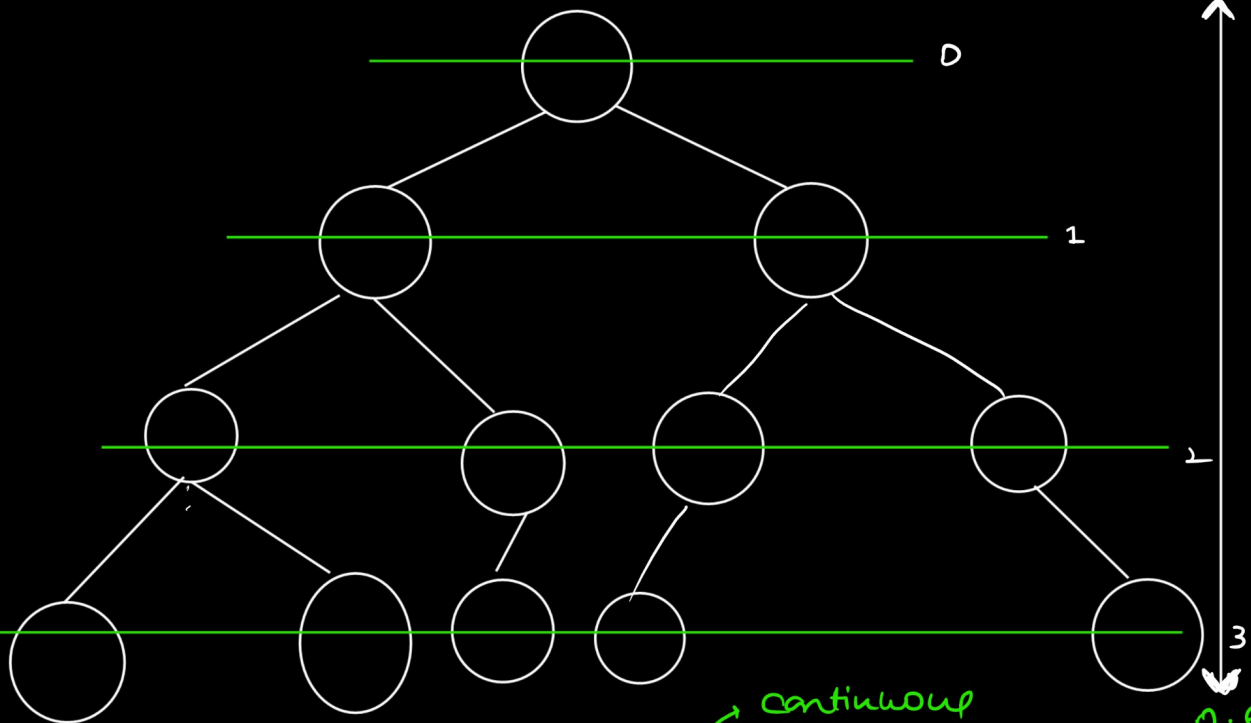
Information Gain ⇒ Entropy(target) - Entropy(class)

how much information each feature is providing to us?

$$\Rightarrow 0.985 - 0.857$$
$$\Rightarrow 0.128$$

Information Gain ⇒ Entropy(target) - Entropy(Gender)

$$\Rightarrow 0.985 - 0.975$$
$$\Rightarrow 0.010$$



Index(['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium',
'total_phenols', 'flavanoids', 'nonflavanoid_phenols',
'proanthocyanins', 'color_intensity', 'hue',
'od280/od315_of_diluted_wines', 'proline'],
dtype='object')

$$1 - \sum_{i=1}^n p_i^2$$

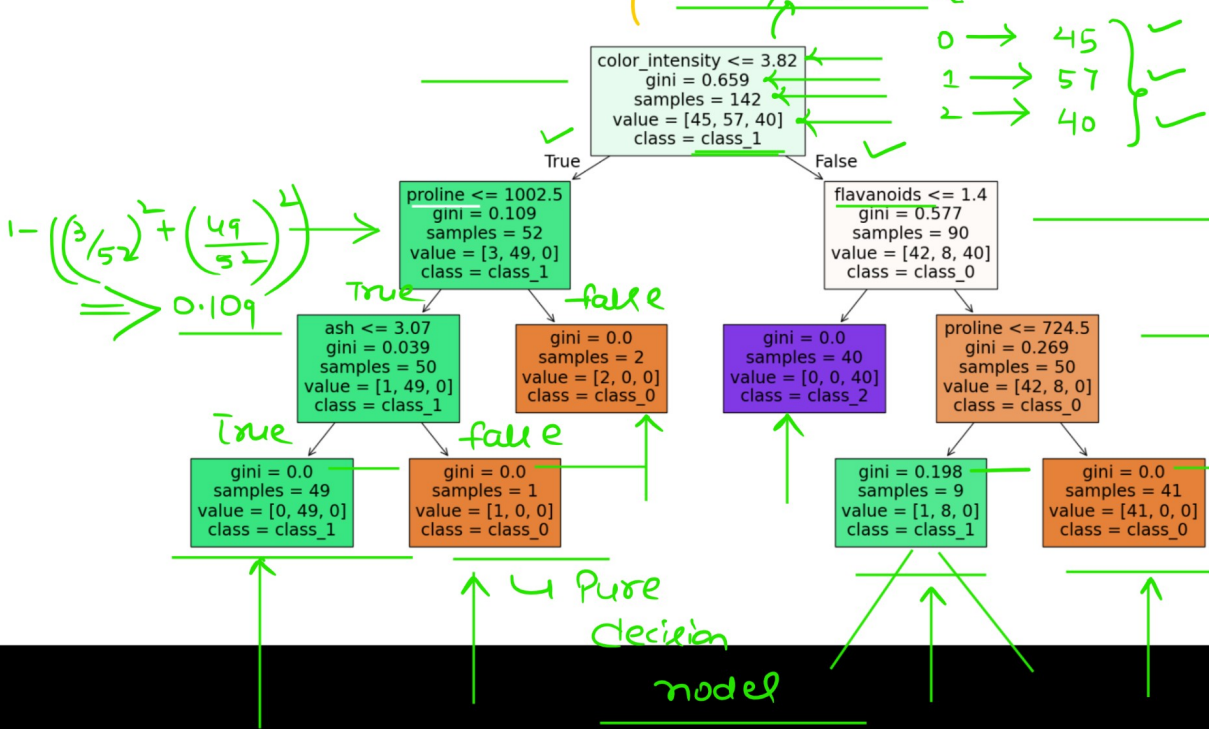
Lower gini value

Decision Tree

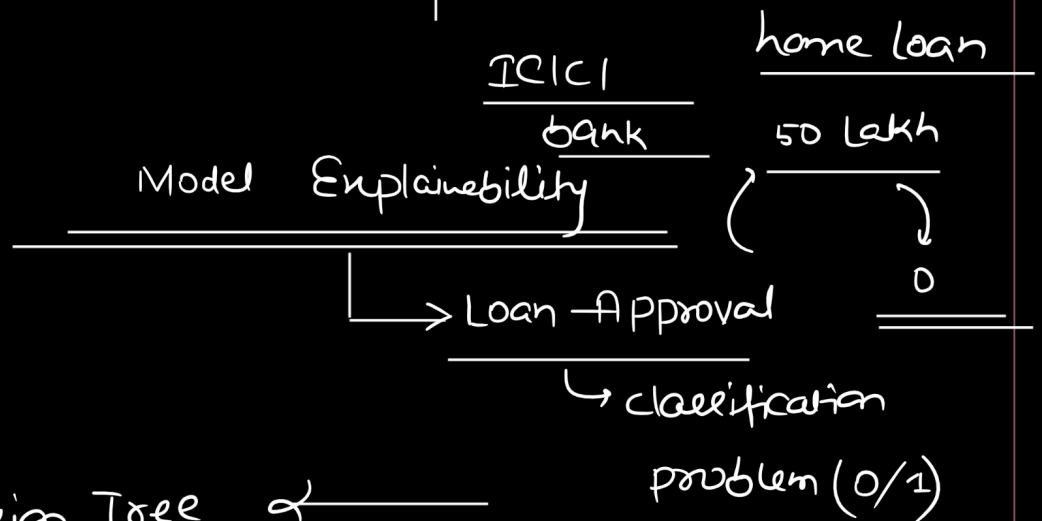
Root Node

mean/median

$$1 - \left(\left(\frac{45}{142} \right)^2 + \left(\frac{57}{142} \right)^2 + \left(\frac{40}{142} \right)^2 \right)$$



	classification based problem	regression based problem
Loss function	Cross, Entropy	MSE



Decision Tree

Model

(0 or 1)

```

graph TD
    A[Decision Tree] --> B[Model]
    B --> C["(0 or 1)"]
  
```

① feature importance

CIBIL score,

salary & many more

```

graph TD
    A["① feature importance"] --> B["CIBIL score, salary & many more"]
  
```

binary classification (0/1) $\rightarrow 2 \times 2$

three classification (0, 1, 2) $\rightarrow 3 \times 3$

		actual	
		0	1
pred	0	TN	FN
	1	FP	TP

		actual		
		0	1	2
pred	0	T0	f0 ✓	f0 ✓
	1	f1 ✓	T1	f1 ✓
	2	f2 ✓	f2 ✓	T2

Pruning $\rightarrow$ Prepruning $\rightarrow$ max_depth = 3
 $\rightarrow$ Postpruning $\rightarrow$ after

successful creation of
decision tree, you can start
deleting the unuseful branches

$x \rightarrow$ test data

